

including CPOE (Computerised Physician Order Entry).

- *Compounded communications:* Electronic documentation, prescription and information exchange allow more efficient communication, and create an audit of all clinical activity at the point of care, and enable internationalisation and language translation (for example, Spanish, French, German, etc).
- *Increased efficiency:* A single repository for clinical information, based on an integrated architecture, provides real-time access to information, eliminating redundant documentation and many manual tasks.
- *Quality improvement:* Technological support for best practices enhances the ability of health centres to comply with performance measures and other health-promotion and disease prevention guidelines.
- *Reduced costs:* Health centres save time and money by curtailing medical errors, eliminating duplicate tests and procedures, automating clinical processes, and limiting the use of paper.

Open MRS

- *Central Concept Dictionary:* All data is stored and defined in a particular format for discrete data, in a centralised dictionary.
- *Patient repository:* Patient data, including demographics, clinical observations, encounter data, orders, etc, is automatically created and maintained.
- *Modular architecture:* An OpenMRS module can extend and add any type of functionality to the existing API and Web app.
- *Localisation/internationalisation:* It has support for multiple languages, extensible to many other languages.
- *Support for complex data:* Supports various unstructured data like radiology images, sound files, etc, which can be stored as 'complex' observations.

Open EMR

- *Electronic billing and insurance:* This increases flexibility and is more convenient for customers, which translates into a greater likelihood of bills being paid on time. With an electronic system, there is no chance of paper cheques getting lost in transit. Thus, payment compliance increases, and late payments are reduced.
- *View and manage patient history:* One can view the patient's record and fix the time for consulting doctors. Health centres can also manage medical insurance claims or generate reports, etc.
- *Electronic prescription:* Prescriptions for medicine can be made on the computer, so paper overheads are reduced.
- *Multi-language support:* The multilingual system makes it easy for patients from different countries to comprehend the facilities and rules.
- *Insurance tracking:* Information on insurance claims

includes MR (Medical Records), DS (Discharge Summary), prescriptions, invoices and other lab test records. It becomes easy to track and assemble all documents within the system itself.

Open EMR supports ANSI X12 EDI billing and two clearing houses: *ProxyMed* (handles healthcare transactions in real-time, such as processing, pre-certifications, eligibility verification and rapid electronic payments) and *ZirMed* (responsible for medical insurance claims).

Design and architecture

OpenVista

It was developed using M—the MUMPS or Massachusetts General Hospital Utility Multi-Programming System—language/database. It currently runs a majority of VistaA systems on the proprietary Intersystem Cache version of MUMPS, and an open source MUMPS database engine, called GT.M for Linux. The OpenVista stack is as follows:

- Client: CareVue (C#, VB, Delphi)
- Network: Web Service (Java)
- Server: OpenVista (M or MUMPS)
- Database Management: MSC Fileman (M) and OVID MEDLINE (Java)
- Application Server: Cache/GT.M (M) and OVID MEDLINE (Java)
- Operating System: Linux
- Hardware: Intel

Open MRS

The top-level architecture of the system consists of:

- The Web layer: With Web applications providing UI (HTML, Javascript)
- The service layer: For business logic and APIs (Java)
- The data access layer: For functions or stored procedures in the database, as well as for access of data (Hibernate, Sql)
- Database: Any relational database
- Modular architecture: It uses a modular architecture to extend the functionality of the system, so that the developer can write new extensions and plug them into the core system.

This project uses many open source tools and technology such as Java, Spring Framework, Spring MVC, Hibernate, MySQL, JQuery, and development tools such as Eclipse, BIRT, Ant, Bamboo and JIRA.

Open EMR

The Open EMR architecture, which uses Linux, Apache, MySQL, PHP or LAMP, consists of four layers:

- The presentation layer: A Web UI for patients (PHP)
- The process layer: For patient admission, scheduling, billing logic (PHP), etc
- The data layer: To operate patient data and for document management (MySQL)